String Loops

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1 Problem

Consider the following problem.

Problem. In a bowl there are n strings of equal length. I repeatedly choose two free ends uniformly at random and tie them together. When all free ends have been tied, some closed string loops are formed. Investigate the number of loops obtained.

In this report I follow the problem-solving structure used in the module. I begin with the Entry Phase, where I identify what is given, what I want to find, and what notation or representations are useful. I then move to the Attack Phase, where I specialise to small cases, make conjectures, describe where I got stuck, and explain the key new idea that led to a full solution. After that, I present the mathematical results in the Solution section, and finally I finish with a Review Phase, where I check the work, reflect on the main breakthrough, and briefly discuss possible extensions.

Throughout the report I write

$$L_n = \text{the number of loops formed from } n \text{ strings.}$$

So the main goal is to understand the probability distribution of L_n, and in particular to find its expectation, variance, and a useful structural description.

2 Entry Phase

2.1 I KNOW

Initially there are n separate strings. Each string has two free ends, so there are $2n$ free ends altogether. At each stage, two free ends are chosen uniformly at random and tied together. The process continues until no free ends remain. At that point, the final object is a disjoint union of closed loops.

A first observation is that the final number of loops must lie between 1 and n. It cannot be 0, because eventually every open chain must close. It cannot exceed n, because there were only n original strings to begin with.

The process is random, so it is natural to think probabilistically. However, the number of possible pairings of the $2n$ endpoints grows quickly, so it is not obvious from the start what the right mathematical model should be.

2.2 I INTRODUCE

I introduce the notation

$$P_n(\ell) = \mathbb{P}(L_n = \ell), \qquad \ell = 1, 2, \ldots, n.$$

I also introduce the idea of computing exact small cases first. Since the problem is combinatorial and probabilistic, this seems like a natural way to build intuition and to produce concrete data that I can later compare against a general formula.

Later in the report I will also introduce another important representation, namely *open chains*, but at the entry stage I did not yet know that this would be the crucial viewpoint.

2.3 I WANT

I want to understand the random variable L_n. More specifically, I want to answer the following questions:

- What is the exact distribution of L_n?
- Can I find a recurrence for the probabilities $P_n(\ell)$?
- What are $\mathbb{E}[L_n]$ and $\mathrm{Var}(L_n)$?
- What kind of distribution is this?
- Can Python simulations support the theoretical results?

At this point I also suspected that the expected number of loops might grow slowly with n, but I did not yet know what the exact formula should be.

2.4 SPECIALISATION

My first serious step was to specialise to the smallest values of n. This helped me avoid making vague guesses, and it also gave me a table of exact values that I could later use to check any conjecture.

Case $n = 1$

This case is immediate. There is only one string, so its two ends must be tied together. Therefore

$$P_1(1) = 1.$$

Case $n = 2$

There are 4 endpoints in total, and there are 3 possible pairings. Of these:

- one pairing ties each string to itself, giving 2 loops;
- the other two pairings link the strings into one larger loop.

Hence

$$P_2(1) = \frac{2}{3}, \qquad P_2(2) = \frac{1}{3}.$$

Already this is interesting: even with only two strings, the most likely outcome is not two small loops but one larger loop.

Case $n = 3$

For $n = 3$, direct case analysis gives

$$P_3(1) = \frac{8}{15}, \qquad P_3(2) = \frac{2}{5}, \qquad P_3(3) = \frac{1}{15}.$$

So the probability of getting the maximum possible number of loops is already quite small.

Cases $n = 4$ and $n = 5$

Continuing in the same way gave

$$P_4(1) = \frac{16}{35}, \qquad P_4(2) = \frac{44}{105}, \qquad P_4(3) = \frac{4}{35}, \qquad P_4(4) = \frac{1}{105},$$

and

$$P_5(1) = \frac{128}{315}, \qquad P_5(2) = \frac{80}{189}, \qquad P_5(3) = \frac{4}{27}, \qquad P_5(4) = \frac{4}{189}, \qquad P_5(5) = \frac{1}{945}.$$

These values are summarised in Table 1.

n	$P_n(1)$	$P_n(2)$	$P_n(3)$	$P_n(4)$	$P_n(5)$
1	1	–	–	–	–
2	$\frac{2}{3}$	$\frac{1}{3}$	–	–	–
3	$\frac{8}{15}$	$\frac{2}{5}$	$\frac{1}{15}$	–	–
4	$\frac{16}{35}$	$\frac{44}{105}$	$\frac{4}{35}$	$\frac{1}{105}$	–
5	$\frac{128}{315}$	$\frac{80}{189}$	$\frac{4}{27}$	$\frac{4}{189}$	$\frac{1}{945}$

Table 1: Exact distributions for L_n for $n = 1, \ldots, 5$.

From this table I observed the following:

- the support of L_n appears to be exactly $\{1, 2, \ldots, n\}$;
- the probability of many loops is small;
- the most likely values remain relatively small even as n increases;
- the expected number of loops appears to grow slowly.

To make that last point more precise, I computed the first few expectations directly:

$$\mathbb{E}[L_1] = 1, \qquad \mathbb{E}[L_2] = \frac{4}{3}, \qquad \mathbb{E}[L_3] = \frac{23}{15},$$

$$\mathbb{E}[L_4] = \frac{176}{105}, \qquad \mathbb{E}[L_5] = \frac{5632}{3150} \approx 1.787.$$

These values increase, but very slowly. This made me suspect that the mean might eventually have logarithmic growth.

3 Attack Phase

3.1 ATTEMPT 1: Direct counting

My first approach was to think directly in terms of the final pairing of the $2n$ endpoints. The number of complete pairings is

$$(2n-1)!! = (2n-1)(2n-3)\cdots 3 \cdot 1.$$

So for small n this is manageable, which is why I could compute the table above. However, the number of pairings grows very quickly, so this does not look like a promising method for obtaining a clean general formula.

The problem was not just the size of the numbers. More importantly, direct counting did not really explain why the probabilities in Table 1 had the particular structure they did. I was producing data, but not understanding.

3.2 CONJECTURE

From the small cases, I began to suspect two things.

First, I conjectured that the number of loops should be built up stage by stage in some simple probabilistic way, rather than requiring a full analysis of final pairings.

Second, I conjectured that the expected number of loops should grow slowly, perhaps logarithmically, because the values in the small table were increasing but not by much.

For a short time, I also wondered whether L_n might behave like a binomial random variable. The process seems to involve repeated yes/no events: either a new loop is created at a given stage or it is not. That made the binomial model feel plausible at first.

3.3 STUCK

However, this binomial intuition did not survive checking. A binomial distribution would require the same success probability at each stage, but there was no reason to expect that here. In fact, the table already suggested that the chance of creating a loop must depend on how many open parts remain.

So this was the point where I became stuck. The small-case evidence was useful, but it was not enough by itself. I needed a better way to describe the process.

The key question became:

What should I track during the process instead of only looking at the end?

3.4 AHA: Open chains

The breakthrough came when I stopped thinking mainly about final loops and instead described the partially tied strings as *open chains*.

Initially, each string is itself an open chain with two free ends, so there are n open chains. Whenever two free ends are tied, exactly one of two things happens:

- two different open chains are joined into one longer open chain;
- the two free ends of the same open chain are tied together, so that chain becomes a closed loop.

In either case, the number of open chains decreases by exactly 1. Therefore the process passes through the states

$$n,\ n-1,\ n-2,\ \ldots,\ 2,\ 1,$$

where the number indicates how many open chains remain.

Once I wrote the problem in this way, the important question became much simpler:

When there are k open chains, what is the probability that the next tie creates a loop?

This felt like the real turning point in the investigation.

3.5 JUSTIFYING the new approach

Suppose there are k open chains remaining. Then there are $2k$ free ends. Fix one free end. It can be tied to any of the other $2k-1$ free ends, all equally likely. Exactly one of those $2k-1$ choices is the other free end of the same chain. If that choice is made, a new loop is formed; otherwise, two different chains are merged.

So at stage k, the probability of creating a loop should be

$$\frac{1}{2k-1}.$$

This was the first moment where the problem started to look fully structured. It suggested that L_n should be describable as a sum of Bernoulli random variables with probabilities

$$1,\ \frac{1}{3},\ \frac{1}{5},\ \ldots,\ \frac{1}{2n-1}.$$

That was my main conjecture at the end of the Attack Phase, and it is exactly what I now prove in the Solution.

4 Solution

4.1 Main theorem

Theorem 1. *For $n \geq 1$,*

$$L_n \stackrel{d}{=} 1 + \sum_{k=2}^{n} X_k,$$

where the random variables $X_2, X_3, \ldots, X_n$ are independent and

$$X_k \sim \text{Bernoulli}\left(\frac{1}{2k-1}\right).$$

Proof. For each $k = 1, 2, \ldots, n$, consider the moment when there are exactly k open chains remaining. Define

$$X_k = \begin{cases} 1, & \text{if the next tie creates a new loop,} \\ 0, & \text{otherwise.} \end{cases}$$

When $k = 1$, the final remaining open chain must close into a loop, so $X_1 = 1$.

Now suppose $k \geq 2$. At this stage there are $2k$ free ends. Fix one free end. It can be tied to any of the other $2k-1$ free ends, all with equal probability. Exactly one of these $2k-1$ endpoints is the other free end of the same open chain. If that particular endpoint is chosen, a loop is created immediately. Any other choice joins two different chains together without creating a loop.

Therefore

$$\mathbb{P}(X_k = 1) = \frac{1}{2k-1}, \qquad \mathbb{P}(X_k = 0) = \frac{2k-2}{2k-1}.$$

The total number of loops formed is exactly the number of stages at which a loop is created. Hence

$$L_n = \sum_{k=1}^{n} X_k = 1 + \sum_{k=2}^{n} X_k.$$

Finally, to justify independence, note that for each stage k, once there are k open chains remaining, the probability that the next tie creates a loop is always

$$\frac{1}{2k-1},$$

regardless of how that state was reached. In other words, for any fixed outcomes of the later-stage variables $X_{k+1}, X_{k+2}, \ldots, X_n$, we still have

$$\mathbb{P}(X_k = 1 \mid X_{k+1}, X_{k+2}, \ldots, X_n) = \frac{1}{2k-1}.$$

So the conditional distribution of X_k does not depend on the earlier history. Repeating this argument for each k shows that the joint distribution factors into the product of the Bernoulli probabilities, and hence $X_2, \ldots, X_n$ are independent. □

This theorem completely explains the structure of the problem. The number of loops is not something determined only at the end. Instead, it is built stage by stage, and at stage k the probability of creating a new loop is simply $1/(2k-1)$.

4.2 Expectation and variance

Corollary 1. *The expectation and variance of L_n are*

$$\mathbb{E}[L_n] = \sum_{k=1}^{n} \frac{1}{2k-1},$$

and

$$\mathrm{Var}(L_n) = \sum_{k=2}^{n} \frac{1}{2k-1}\left(1 - \frac{1}{2k-1}\right).$$

Proof. These formulas follow immediately from the theorem, because L_n is a sum of independent Bernoulli random variables. □

This also shows that L_n is not binomial. More precisely, it is a shifted *Poisson-binomial* random variable:

$$L_n = 1 + \sum_{k=2}^{n} X_k, \qquad X_k \sim \mathrm{Bernoulli}\left(\frac{1}{2k-1}\right).$$

The expectation formula matches the small cases exactly:

$$\mathbb{E}[L_1] = 1,$$

$$\mathbb{E}[L_2] = 1 + \frac{1}{3} = \frac{4}{3},$$

$$\mathbb{E}[L_3] = 1 + \frac{1}{3} + \frac{1}{5} = \frac{23}{15},$$

$$\mathbb{E}[L_4] = 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} = \frac{176}{105},$$

$$\mathbb{E}[L_5] = 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9} \approx 1.787302.$$

So the theorem is completely consistent with the earlier table.

4.3 A recurrence for the exact distribution

Proposition 1. *For $n \geq 2$ and $1 \leq \ell \leq n$,*

$$P_n(\ell) = \frac{2n-2}{2n-1} P_{n-1}(\ell) + \frac{1}{2n-1} P_{n-1}(\ell - 1),$$

with initial condition $P_1(1) = 1$, and with the convention that probabilities outside the valid range are zero.

Proof. From the theorem,

$$L_n = L_{n-1} + X_n,$$

where $X_n \sim \mathrm{Bernoulli}\left(\frac{1}{2n-1}\right)$ and X_n is independent of L_{n-1}. Therefore

$$\mathbb{P}(L_n = \ell) = \mathbb{P}(L_{n-1} = \ell, X_n = 0) + \mathbb{P}(L_{n-1} = \ell - 1, X_n = 1),$$

which immediately gives the recurrence. □

This recurrence is useful because it turns the probabilistic description into an exact computational rule. In practice, it is much easier to generate the distribution this way than to count all pairings directly.

For example,

$$P_3(1) = \frac{4}{5}P_2(1) = \frac{4}{5} \cdot \frac{2}{3} = \frac{8}{15},$$

and

$$P_3(2) = \frac{4}{5}P_2(2) + \frac{1}{5}P_2(1) = \frac{4}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{2}{3} = \frac{2}{5}.$$

So the recurrence reproduces the small-case values found earlier by hand.

4.4 Further consequences

Because L_n is a sum of independent Bernoulli random variables, its probability generating function is

$$G_n(t) = \mathbb{E}[t^{L_n}] = t \prod_{k=2}^{n} \left(1 + \frac{t-1}{2k-1}\right).$$

The expectation can also be rewritten in terms of harmonic numbers:

$$\mathbb{E}[L_n] = \sum_{k=1}^{n} \frac{1}{2k-1} = H_{2n} - \frac{1}{2}H_n,$$

where $H_n = \sum_{j=1}^{n} \frac{1}{j}$.

Using the standard asymptotic formula for harmonic numbers, this gives

$$\mathbb{E}[L_n] \sim \frac{1}{2}\log n + \log 2 + \frac{\gamma}{2} \qquad \text{as } n \to \infty,$$

where γ is Euler's constant.

This confirms the pattern I suspected from the small cases: the expected number of loops does grow with n, but only very slowly.

The theorem also makes it easy to compute two extreme probabilities. To get exactly one loop, no loop can be created at any stage $n, n-1, \ldots, 2$, so

$$\mathbb{P}(L_n = 1) = \prod_{k=2}^{n} \frac{2k-2}{2k-1}.$$

To get the maximum possible number of loops, namely n, a loop must be formed at every stage, so

$$\mathbb{P}(L_n = n) = \prod_{k=2}^{n} \frac{1}{2k-1}.$$

These formulas agree with the first and last entries of Table 1.

4.5 Python checks and numerical evidence

After proving the theoretical results, I used Python to check them numerically. I wanted the code to do more than produce one small example. In particular, I wanted to test whether the exact formulas continued to agree with simulation for larger values of n.

Code representation

In the code, each original string begins as `[1,1]`. Here a 1 represents a free end and a 0 represents a part that has already been tied.

For example:

- two separate strings `[1,1]` and `[1,1]` can be joined to make `[1,0,0,1]`;
- if the two remaining free ends of a single chain are tied together, the chain is recorded as a closed loop, that is, a list containing only zeros.

I organised the code around three main functions:

1. `simulate_once(n)`, which runs one random experiment;
2. `estimate_distribution(n, trials)`, which estimates the distribution from repeated simulation;
3. `plot_distribution(n, trials)`, which compares the simulated and exact distributions graphically.

The code used in this report is available in the following GitHub repository:

https://github.com/r00000nn/MA3K7-Problem-Solving-with-Python

First numerical check: $n = 5$

I first checked the exact distribution when $n = 5$. From the recurrence, the exact distribution is

$$\{1 : 0.406349,\ 2 : 0.423280,\ 3 : 0.148148,\ 4 : 0.021164,\ 5 : 0.001058\}.$$

A simulation with 10,000 trials gave

$$\{1 : 0.4066,\ 2 : 0.4212,\ 3 : 0.1473,\ 4 : 0.0232,\ 5 : 0.0017\}.$$

These are already quite close. This was a useful first check, but since $n = 5$ is still small, I wanted to test the formulas more seriously.

Larger values of n

I also simulated the process for

$$n = 10,\ 20,\ 50,\ 100.$$

For each value I compared:

- the simulated mean;
- the theoretical mean $\sum_{k=1}^{n} \frac{1}{2k-1}$;
- the asymptotic approximation $\frac{1}{2}\log n + \log 2 + \frac{\gamma}{2}$;
- the simulated variance;
- the theoretical variance $\sum_{k=2}^{n} \frac{1}{2k-1}\left(1 - \frac{1}{2k-1}\right)$.

The results are shown in Table 2.

n	Simulated mean	Theoretical mean	Asymptotic mean	Simulated variance	Theoretical variance
5	1.790200	1.787302	1.786474	0.612384	0.603437
10	2.143600	2.133256	2.133048	0.934179	0.924534
20	2.489200	2.479673	2.479621	1.282083	1.258470
50	2.952600	2.937775	2.937767	1.736953	1.709074
100	3.292000	3.284342	3.284340	2.069136	2.053142

Table 2: Comparison of simulated and theoretical moments using 10,000 trials.

What I found especially striking is how accurate the asymptotic formula already is. Even at $n = 10$, the asymptotic mean is extremely close to the exact mean, and by $n = 50$ and $n = 100$ the agreement is almost indistinguishable at the precision shown.

This strongly supports the conclusion that the expected number of loops grows logarithmically.

Comparing the full distributions

Checking only the mean and variance is useful, but it does not show by itself that the whole simulated distribution matches the exact theory. To examine this more carefully, I compared the exact and simulated distributions directly using two error measures:

- the maximum absolute difference between corresponding probabilities;
- the total variation distance.

The results are shown in Table 3.

n	Max absolute error	Total variation distance
5	0.010048	0.010506
10	0.005340	0.008088
20	0.003791	0.004935
50	0.002181	0.004855

Table 3: Distance between simulated and exact distributions using 10,000 trials.

These values are all small. So the simulation does not only reproduce the first few moments; it matches the full distribution quite well.

Plots of the distributions

I also plotted the simulated and exact distributions for $n = 10$, $n = 20$, and $n = 50$. These figures reveal the overall shape of the distribution more clearly than a table alone.

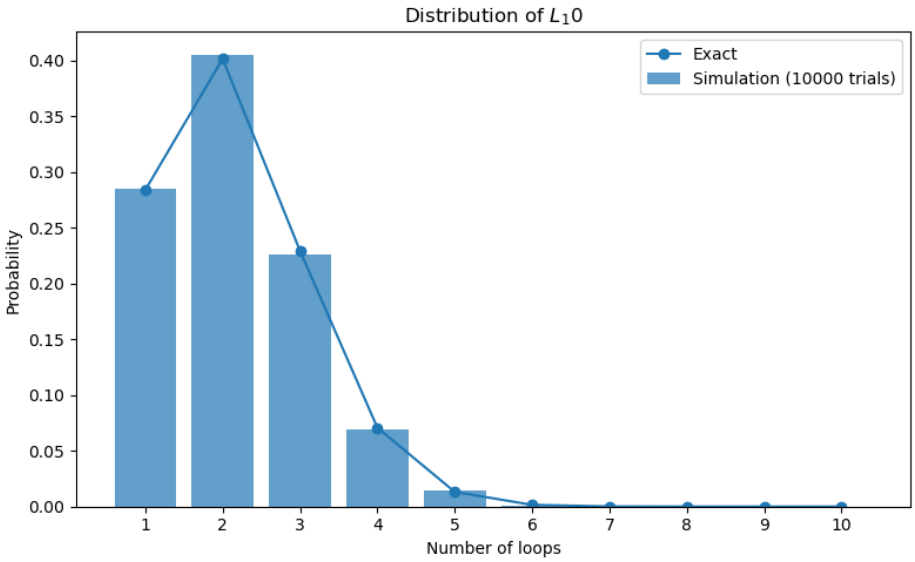

Figure 1: Simulated and exact distributions for L_{10}. Most of the probability mass is concentrated on 1, 2, and 3 loops, with the largest mass around 2 loops.

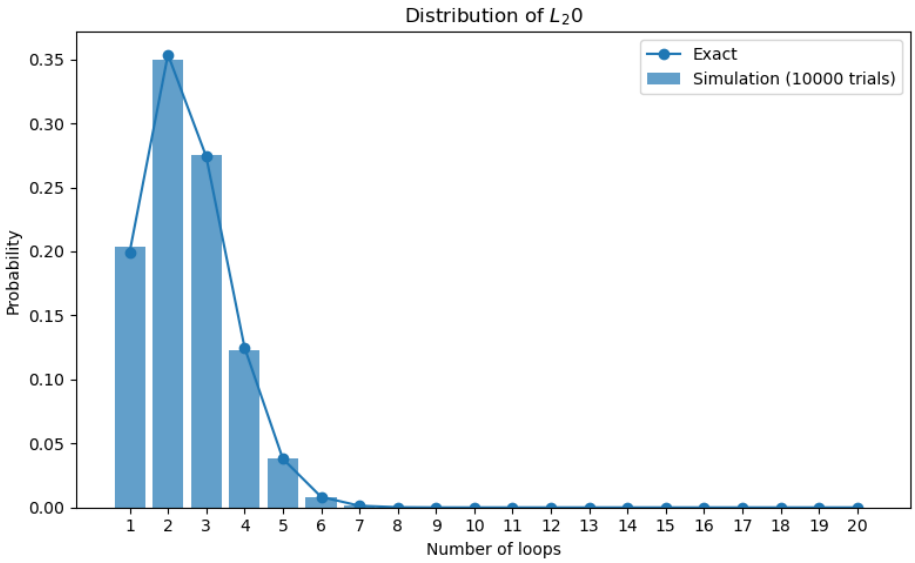

Figure 2: Simulated and exact distributions for L_{20}. The distribution has shifted slightly to the right, but it is still concentrated on small values.

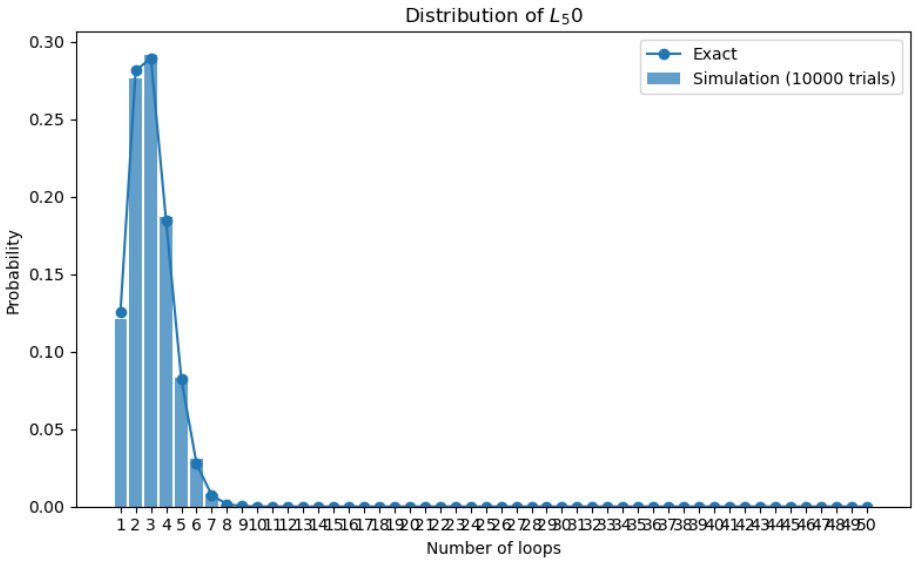

Figure 3: Simulated and exact distributions for L_{50}. Even when starting with 50 strings, the number of loops is usually still only around 2, 3, or 4, which is consistent with logarithmic growth of the mean.

These figures helped me see an aspect of the problem that is easy to miss if one only stares at formulas: although the maximum possible number of loops grows with n, the actual distribution remains heavily concentrated near the left-hand side for a long time.

5 Review Phase

5.1 CHECK

I checked my work in several different ways:

- by computing small exact cases directly;
- by checking that the recurrence reproduces those small cases;
- by comparing simulated and exact distributions;
- by comparing simulated and theoretical means and variances;
- by comparing the exact mean with its asymptotic approximation.

This mattered because I did not want the final result to rest on only one line of reasoning. The small-case calculations, recurrence, theorem, and Python simulations all support the same conclusion from different directions.

5.2 REFLECT

The most important moment in the investigation was realising that direct counting of final pairings was not the right perspective. Initially I thought that if I pushed the counting far enough, a pattern would reveal itself. But this was not really happening. What I was getting was more data, not more understanding.

The real progress came from changing the object I was tracking. Once I started thinking in terms of open chains rather than final loops, the process became much simpler. At stage k, there are $2k$ free ends, and only one of the $2k-1$ possible partners closes the chosen chain.

From that point on, the central probability $1/(2k-1)$ becomes obvious, and the rest of the theory follows naturally.

So the main lesson I take from this problem is that reformulation is often the decisive step in problem solving. In this case, the breakthrough was not a complicated combinatorial trick, but a better choice of viewpoint.

This problem also highlighted a useful probabilistic idea: not every sum of Bernoulli-type events is binomial. Here the random variable does arise from repeated yes/no events, but the probabilities change from stage to stage, so the correct model is Poisson-binomial rather than binomial.

I found this a helpful reminder that recognising the right distributional structure matters just as much as performing calculations.

5.3 EXTEND OR GENERALISE

There are several natural ways the problem could be extended.

- The endpoints could be chosen with a non-uniform rule rather than uniformly at random.
- Some pairings could be forbidden.
- The strings could be labelled or oriented.
- One could ask for a limiting distribution after suitable centring and scaling.

I did not pursue these here, but they suggest that the method developed in this report could be adapted to a wider class of random linking problems.

Overall, this investigation moved through all the stages of the rubric in a natural way: I began by identifying what was given and by specialising to small cases, then I attacked the problem by forming and revising conjectures, got stuck, found a more useful point of view, and finally justified the resulting mathematics and checked it computationally. For me, the most valuable part of the whole process was seeing how a simple change of viewpoint could turn a messy counting problem into a clean probabilistic structure.